

ZACHARY DIAKS

Senior Quality Engineer

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EDUCATION

Northeastern University — Boston, MA
B.S. Physics, Magna Cum Laude (May 2020)
Minors: Mathematics and Computer Science

TECHNOLOGIES

Languages: MATLAB, Python, JavaScript, Simulink, C++, Bash, SQL
Frameworks: MUnit (MATLAB), QUnit, Playwright, Sinon, Jest, Pytest, React

WORK EXPERIENCE

MATHWORKS

Senior Quality Engineer (May 2026 – Present)

- Head Quality Engineer for the Sensor Fusion and Tracking Toolbox, owning test strategy and quality. Collaborate with developers to plan and implement automated test suites for MATLAB (MUnit), JavaScript (Sinon, Playwright, QUnit), and Agentic AI (evaluation suites with LLM-as-a-Judge) features.
- Deliver quarterly product quality retrospectives, analyzing KPIs like market bug trends, code complexity, and user experience surveys and telemetry.
- Serve as a Quality Engineering representative in a cross-department working group aiming to redefine the standards and guidelines for agentic software development processes.
- Develop a multi-agent skill to plan, implement, and review standards-compliant tests from software specification documents, saving pilot participants across multiple departments up to one day of test planning, writing, and qualification.
- Design multi-agent skill to automate the generation of quality retrospective reports, connecting data sources from multiple areas of the company. Anecdotes from pilot teams cite time savings of up to 8 hours per report generation.

Quality Engineer III (May 2023 – May 2026)

- Created a MATLAB-based prompt evaluation framework for regression testing, leveraging static analysis and benchmark responses for grading. Framework was used by internal automation tools which leveraged LLMs.
- Optimized default test selection strategy for internal build and test system by leveraging source to test mapping, saving as much as 100 hours of linear test time per job.
- Analyzed UX data to identify frequently hit and untested errors to improve both test coverage and usability.

Quality Engineer I (October 2021 – May 2023)

- Contributed to an internal tool used to build test bed reports from markup in test files (MATLAB backend, React frontend, MySQL database).

Application Support Engineer (June 2020 – October 2021)

- Provided technical support to customers for MathWorks products in the Deployment and Test and Measurement areas (e.g. MATLAB Compiler, Database Toolbox, Data Acquisition Toolbox).

ROGERS SCIENCES | *R&D Engineering Co-op (July 2019 – December 2019)*

- Designed test fixtures to validate device specifications against average patient data in FDA databases to prepare for FDA approval.
- Automated device calibration tasks using C++, saving minutes of effort per calibration.

HARVARD MEDICAL SCHOOL | *Neuroimaging Co-op (July 2018 – May 2019)*

- Operated Transmission Electron Microscope and prepared images for 3D reconstruction using Bash/Python.
- Automated image processing tasks using Slack API, reducing manual effort from several hours to less than a minute.